

HONG KONG OBSERVATORY, Hong Kong

WMO: 450050

Lat: 22.302N

Lon: 114.174E

Elev: 203

StdP: 14.59

Time Zone: 8.00 (E08)

Period: 82-92

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 49.3       | 51.6 | 30.2                          | 24.5 | 55.1 | 35.3 | 30.4 | 57.2 | 20.6                  | 60.6 | 18.9 | 60.4 | 5.0                   | 10   | 0.365 |

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                     | (p)  |
| 7             | 6.3                       | 90.0            | 79.7 | 89.0 | 79.6 | 88.1 | 79.4 | 81.3                | 86.8 | 80.9 | 86.3 | 80.5 | 85.8 | 7.6                     | 270  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 79.9                            | 156.2 | 84.7 | 79.2 | 152.9 | 84.4 | 79.0 | 151.4 | 84.2 | 45.3          | 87.1 | 44.8 | 86.6 | 44.4 | 86.2 | 83.1              |

Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|--|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |  |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |  |
| 19.3              | 16.6 | 14.5 | DB                         | 45.7 | 92.3               | 3.1 | 0.6                                                | 43.4 | 92.7       | 41.6 | 93.1       | 39.8 | 93.4       | 37.6 | 93.8 |  |
|                   |      |      | WB                         | 39.2 | 82.7               | 3.2 | 0.3                                                | 36.9 | 82.9       | 35.0 | 83.0       | 33.2 | 83.2       | 30.9 | 83.4 |  |

Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 73.6   | 61.1  | 60.9  | 65.3  | 71.4  | 78.4  | 82.3  | 84.2  | 83.8  | 82.0  | 77.9  | 71.0  | 63.7  |
|                                                                                  | DBStd       | 9.53   | 4.27  | 4.62  | 5.99  | 4.86  | 3.61  | 3.06  | 2.33  | 2.29  | 2.70  | 3.33  | 4.49  | 5.13  |
|                                                                                  | HDD50       | 3      | 1     | 0     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     |
|                                                                                  | HDD65       | 426    | 130   | 128   | 71    | 6     | 0     | 0     | 0     | 0     | 0     | 0     | 8     | 83    |
|                                                                                  | CDD50       | 8607   | 346   | 307   | 476   | 643   | 880   | 969   | 1059  | 1046  | 961   | 865   | 630   | 425   |
|                                                                                  | CDD65       | 3556   | 10    | 14    | 82    | 199   | 415   | 519   | 594   | 581   | 511   | 401   | 188   | 42    |
|                                                                                  | CDH74       | 33513  | 0     | 2     | 143   | 784   | 3286  | 5807  | 7398  | 7064  | 5627  | 2982  | 411   | 9     |
| CDH80                                                                            | 10535       | 0      | 0     | 0     | 55    | 658   | 1945  | 3052  | 2738  | 1722  | 360   | 5     | 0     |       |
| Wind                                                                             | WSAvg       | 7.4    | 7.1   | 8.3   | 8.0   | 7.9   | 7.5   | 7.4   | 7.0   | 6.5   | 7.2   | 8.1   | 7.0   | 6.6   |
| Precipitation                                                                    | PrecAvg     | 87.6   | 0.9   | 1.9   | 2.7   | 6.3   | 12.5  | 15.1  | 13.3  | 15.2  | 11.8  | 5.4   | 1.5   | 1.2   |
|                                                                                  | PrecMax     | 127.9  | 4.0   | 9.5   | 16.9  | 19.4  | 30.4  | 37.9  | 45.2  | 34.3  | 31.4  | 28.3  | 5.7   | 8.1   |
|                                                                                  | PrecMin     | 35.5   | 0.0   | 0.0   | 0.0   | 0.2   | 0.2   | 4.2   | 4.1   | 3.3   | 0.9   | 0.0   | 0.0   | 0.0   |
|                                                                                  | PrecStd     | 20.2   | 0.9   | 2.4   | 3.3   | 4.6   | 8.0   | 7.4   | 8.0   | 8.7   | 7.4   | 6.7   | 1.7   | 2.0   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 71.8  | 73.7  | 78.8  | 83.5  | 88.2  | 90.0  | 91.4  | 91.2  | 90.5  | 87.0  | 80.5  | 75.3  |
|                                                                                  |             | MCWB   | 65.3  | 67.6  | 72.8  | 75.7  | 79.2  | 79.8  | 80.3  | 79.8  | 78.2  | 77.3  | 71.5  | 66.8  |
|                                                                                  | 2%          | DB     | 69.7  | 71.5  | 77.1  | 81.7  | 86.4  | 88.7  | 90.0  | 89.8  | 88.6  | 84.7  | 78.7  | 72.7  |
|                                                                                  |             | MCWB   | 63.6  | 66.8  | 72.7  | 75.6  | 78.8  | 79.8  | 80.1  | 79.8  | 78.4  | 76.3  | 71.0  | 64.5  |
|                                                                                  | 5%          | DB     | 68.0  | 69.2  | 75.7  | 80.0  | 84.9  | 87.6  | 88.9  | 88.8  | 87.4  | 83.2  | 77.4  | 71.1  |
|                                                                                  |             | MCWB   | 62.5  | 65.2  | 72.0  | 74.9  | 78.3  | 79.8  | 79.9  | 79.6  | 78.2  | 75.5  | 70.6  | 63.7  |
|                                                                                  | 10%         | DB     | 66.4  | 67.2  | 73.9  | 78.7  | 83.5  | 86.5  | 87.9  | 87.7  | 86.2  | 81.9  | 76.4  | 69.8  |
|                                                                                  |             | MCWB   | 61.2  | 63.4  | 70.5  | 74.5  | 77.7  | 79.6  | 79.6  | 79.5  | 77.9  | 74.8  | 70.0  | 63.0  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 67.2  | 69.9  | 74.5  | 77.6  | 80.4  | 81.8  | 81.9  | 81.9  | 81.2  | 79.2  | 75.0  | 68.2  |
|                                                                                  |             | MCDB   | 69.9  | 72.3  | 77.3  | 80.8  | 85.6  | 87.0  | 87.7  | 88.0  | 86.6  | 83.9  | 78.1  | 72.7  |
|                                                                                  | 2%          | WB     | 65.3  | 67.8  | 73.6  | 76.4  | 79.6  | 81.0  | 81.1  | 81.0  | 80.5  | 78.1  | 73.4  | 66.9  |
|                                                                                  |             | MCDB   | 68.4  | 70.7  | 76.5  | 79.8  | 84.7  | 86.0  | 86.8  | 86.6  | 85.3  | 82.6  | 77.0  | 71.2  |
|                                                                                  | 5%          | WB     | 63.6  | 66.1  | 72.5  | 75.5  | 79.0  | 80.7  | 80.8  | 80.7  | 79.9  | 77.3  | 72.3  | 65.4  |
|                                                                                  |             | MCDB   | 67.0  | 68.9  | 75.4  | 79.3  | 83.9  | 85.6  | 86.5  | 86.0  | 84.7  | 81.7  | 76.3  | 69.9  |
|                                                                                  | 10%         | WB     | 62.2  | 64.4  | 70.7  | 74.8  | 78.3  | 80.1  | 80.4  | 80.1  | 79.2  | 76.3  | 71.3  | 64.1  |
|                                                                                  |             | MCDB   | 65.7  | 66.6  | 73.6  | 78.5  | 82.8  | 84.9  | 86.0  | 85.4  | 84.2  | 80.9  | 75.6  | 68.7  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 6.3   | 5.6   | 6.1   | 6.0   | 5.9   | 5.4   | 6.3   | 6.4   | 6.0   | 5.7   | 6.4   | 6.9   |
|                                                                                  |             | MCDBR  | 7.5   | 8.2   | 7.7   | 7.6   | 7.2   | 6.5   | 7.3   | 7.7   | 7.6   | 6.7   | 6.5   | 7.4   |
|                                                                                  |             | MCWBR  | 4.8   | 5.6   | 4.6   | 3.9   | 3.1   | 2.4   | 2.3   | 2.9   | 3.3   | 3.4   | 3.9   | 4.4   |
|                                                                                  | 5% WB       | MCDBR  | 6.4   | 7.6   | 7.3   | 7.1   | 6.4   | 5.9   | 6.5   | 6.9   | 6.6   | 5.7   | 5.6   | 6.5   |
| MCWBR                                                                            |             | 4.8    | 5.7   | 5.3   | 4.3   | 3.4   | 2.6   | 2.7   | 3.0   | 3.0   | 2.7   | 3.9   | 4.7   |       |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | 0.576 | 0.627 | 0.751 | 0.728 | 0.567 | 0.536 | 0.525 | 0.590 | 0.623 | 0.668 | 0.570 | 0.564 |
|                                                                                  | taud        |        | 1.814 | 1.742 | 1.537 | 1.622 | 2.019 | 2.121 | 2.149 | 1.946 | 1.845 | 1.697 | 1.895 | 1.865 |
|                                                                                  | Ebn at Noon |        | 217   | 216   | 197   | 206   | 240   | 246   | 248   | 233   | 223   | 203   | 217   | 214   |
|                                                                                  | Edh at Noon |        | 62    | 71    | 90    | 84    | 56    | 50    | 49    | 60    | 65    | 73    | 57    | 57    |
| All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 961   | 987   | 1073  | 1208  | 1440  | 1478  | 1710  | 1596  | 1503  | 1392  | 1120  | 999   |
|                                                                                  | RadStd      |        | 159   | 159   | 143   | 179   | 189   | 219   | 180   | 145   | 118   | 139   | 128   | 147   |

Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (13 neighbors) | N/A   | N/A     | N/A    | +0.45   | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page